



兰州工业学院

实 验 报 告

(2022 — 2023 学年第 二 学期)

课程名称 《计算机网络应用》

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实验项目	实验五 NAT 地址转换		
实验时间	2024 年 6 月 3 日	实验地点	信息与通信工程基础实验室 (EDA 技术实验室)

一、实验目的

- 1.掌握配置 NAT 的 3 种方法；
- 2.掌握静态 NAT 的配置方法；

二、实验环境：

Cisco Packet Tracer 模拟软件

三、实验步骤

1、按如图所示搭建实验环境，配置网络参数。拓扑图中的路由器设置时钟速率为 12800。

本实验要求通过对静态路由、静态 NAT，实现拓扑中全网的连通性。

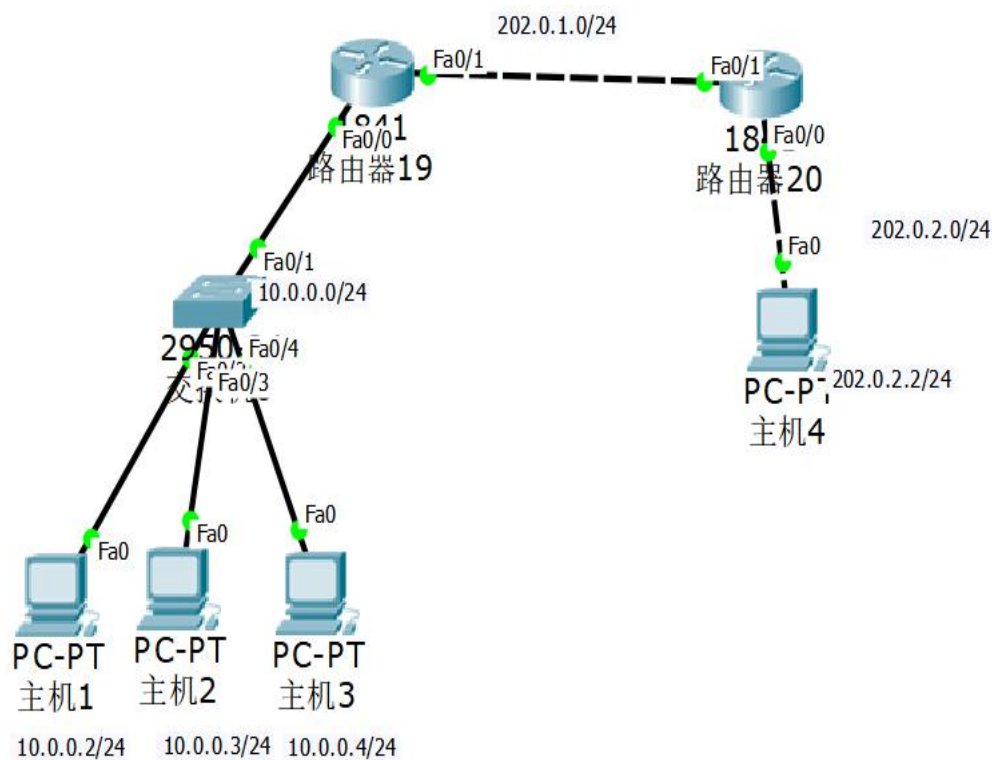


图 1 实验拓扑结构图

在 Cisco Packet Tracer 模拟软件的拓扑图

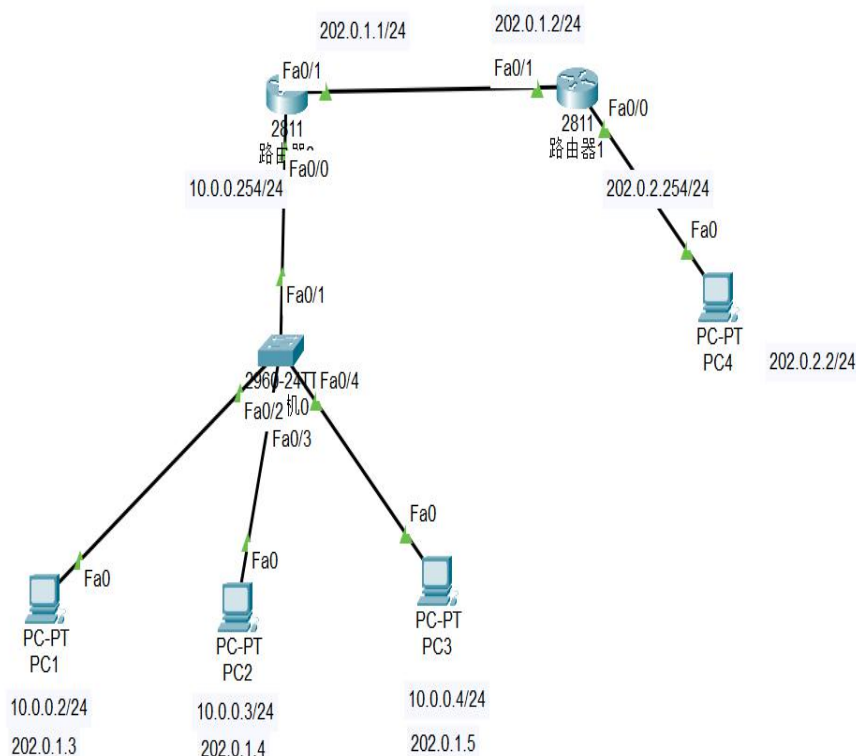


图 1 拓扑图

PC1、PC2（最上面两台计算机）连接到
路由器相应端口的信息（端口号及相应
地址）

FastEthernet0/0
10.0.0.254/24

路由器的端口信息
（端口号及相应地址）

路由器 0: FastEthernet0/0, 10.0.0.254/24
FastEthernet0/1, 202.0.1.1/24
路由器 1: FastEthernet0/0, 202.0.2.254/24
FastEthernet0/1, 202.0.1.2/24

PC1、PC2 的 IP 地址、子网掩码
及网关信息

PC1: 10.0.0.2/24 10.0.0.254
PC2: 10.0.0.3/24 10.0.0.254
PC3: 10.0.0.4/24 10.0.0.254

配置前 PC 之间的连通性测试结果图

```
C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=13ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

图 2 PC1 测试和其他 PC 的连通性

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=14ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

图 3 PC2 测试和其他 PC 的连通性

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=9ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 4 PC3 测试和其他 PC 的连通性

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 5 PC4 测试和其他 PC 的连通性

静态 NAT 命令

```
Router(config)#int fa0/0
Router(config-if)#ip nat ins
Router(config-if)#ip nat inside
Router(config-if)#exi
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exi
Router(config)#ip nat inside source static 10.0.0.2 202.0.1.3
Router(config)#ip nat inside source static 10.0.0.3 202.0.1.4
Router(config)#ip nat inside source static 10.0.0.4 202.0.1.5
Router(config)#
```

复制

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图 6 路由器 0 的静态 NAT 命令

```
Router#show ip nat t
Pro  Inside global      Inside local      Outside local     Outside global
icmp 202.0.1.3:1         10.0.0.2:1       202.0.2.2:1      202.0.2.2:1
icmp 202.0.1.3:2         10.0.0.2:2       202.0.2.2:2      202.0.2.2:2
icmp 202.0.1.3:3         10.0.0.2:3       202.0.2.2:3      202.0.2.2:3
icmp 202.0.1.3:4         10.0.0.2:4       202.0.2.2:4      202.0.2.2:4
icmp 202.0.1.3:5         10.0.0.2:5       202.0.2.2:5      202.0.2.2:5
icmp 202.0.1.3:6         10.0.0.2:6       202.0.2.2:6      202.0.2.2:6
icmp 202.0.1.3:7         10.0.0.2:7       202.0.2.2:7      202.0.2.2:7
icmp 202.0.1.3:8         10.0.0.2:8       202.0.2.2:8      202.0.2.2:8
--- 202.0.1.3           10.0.0.2         ---              ---
--- 202.0.1.4           10.0.0.3         ---              ---
--- 202.0.1.5           10.0.0.4         ---              ---
Router#
```

复制

粘贴

图 7 PC 机私有 IP 地址映射为公共 IP 地址

静态路由命令

```
Router(config-if)#ip route 202.0.2.0 255.255.255.0 202.0.1.2
Router(config)#ip route 202.0.2.0 255.255.255.0 202.0.1.2
Router(config)#int fa0/0
Router(config-if)#ip nat ins
```

图 8 路由器 0 的静态路由命令

```
Router(config)#ip route 10.0.0.0 255.255.255.0 202.0.1.1
Router(config)#
```

图 9 路由器 0 的静态路由命令

配置后 PC 之间的连通性测试结果图


```

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=6ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

图 9 PC1 与其他 PC 的连通性测试

```

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

图 10 PC2 与其他 PC 的连通性测试

```

C:\> ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\> ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\> ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time=1ms TTL=126
Reply from 202.0.2.2: bytes=32 time=7ms TTL=126
Reply from 202.0.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

```

图 11 PC3 与其他 PC 的连通性测试

```

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 202.0.1.3: bytes=32 time<1ms TTL=126
Reply from 202.0.1.3: bytes=32 time<1ms TTL=126
Reply from 202.0.1.3: bytes=32 time<1ms TTL=126
Reply from 202.0.1.3: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 202.0.1.4: bytes=32 time<1ms TTL=126
Reply from 202.0.1.4: bytes=32 time<1ms TTL=126
Reply from 202.0.1.4: bytes=32 time<1ms TTL=126
Reply from 202.0.1.4: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 202.0.1.5: bytes=32 time<1ms TTL=126
Reply from 202.0.1.5: bytes=32 time<1ms TTL=126
Reply from 202.0.1.5: bytes=32 time<1ms TTL=126
Reply from 202.0.1.5: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

图 12 PC4 与其他 PC 的连通性测试

四、总结

本次实验通过配置静态 NAT 和静态路由，实现了局域网内部及外部网络的正常通信。实验中使用 Cisco Packet Tracer 搭建网络环境，详细配置了各设备的 IP 地址、子网掩码及网关信息。通过静态 NAT 命令和静态路由命令，成功实现了各 PC 之间的互通，验证了配置的正确性和网络的连通性。

通过本次实验，我熟练掌握了 NAT 的配置方法，特别是静态 NAT 的配置。静态 NAT 是 NAT 的一种基本形式，通过一对一地将内网地址映射为外网地址，实现了内外网的通信。实验不仅加深了我对 NAT 技术在实际网络环境中的应用理解，还提高了对复杂网络配置的处理能力。我在实验中学会了如何在实际操作中应用理论知识，解决实际问题。

本实验为将来的网络设计和维护打下了坚实的基础。通过本次实验的训练，我具备了在真实环境中配置和管理网络的能力，能够有效应对各种网络配置和维护的挑战，为未来的职业发展提供了有力支持。

实验成绩评定表

序号	考核项目	分值分布	成绩
1	出勤与纪律情况	10	
2	实验任务完成情况	40	
3	实验报告质量	50	
总分			
指导教师签字			

备注：考核项目及分值分布应依据教学大纲确定。